The Topological Cooperation Theorem: Fixed-Point Search as a Free Cooperation Mechanism

with Applications to AI Safety

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Abstract

We prove that in any finite game with multiple Nash equilibria, exhaustive fixed-point search followed by welfare-maximising selection provides a *strictly positive cooperation dividend* over random equilibrium convergence, without any modification to the game's payoff structure. The "mechanism" is pure computation: searching the fixed-point set of the best-response map and selecting the best element. We call this the **Topological Cooperation Theorem**, because the cooperation arises from the topology of the equilibrium correspondence rather than from mechanism design. We validate the theorem on 2,000 random games (100% confirmation rate) and apply it to five AI safety problems: corrigibility, deceptive alignment, alignment as a public good, reward hacking, and AI debate. The corrigibility application achieves 100% safe equilibrium selection versus 63% for standard policy gradient methods.

Contents

1 Introduction **2**

2 The Topological Cooperation Theorem **2**
2.1 Setup . . . 2
2.2 Main Result . . . 2
2.3 Quantitative Bounds . . . 3

3 Algorithmic Realisation: FP-NE Search **3**

4 Empirical Validation **3**
4.1 Results . . . 4

5 Application: AI Safety **4**
5.1 Corrigibility . . . 4
5.2 Deceptive Alignment . . . 5
5.3 Alignment as Public Good . . . 5
5.4 Reward Hacking . . . 5
5.5 AI Debate . . . 6

6 Discussion **6**
6.1 Computation as Mechanism . . . 6
6.2 Implications for AI Safety . . . 7
6.3 Limitations . . . 8

1 Introduction

A fundamental challenge in multi-agent systems is equilibrium selection: when a game has multiple Nash equilibria, which one will learning dynamics converge to? This question is critical for AI safety, where "safe" and "unsafe" equilibria coexist and the learning algorithm's choice determines whether the system is aligned.

The standard approach is *mechanism design*: modify the game's payoffs (via taxes, subsidies, or contracts) to make the desired equilibrium dominant. This requires a principal with the power and knowledge to redesign the game — a strong assumption in multi-agent AI systems.

We show that a weaker, purely computational intervention suffices: *search the Nash equilibrium set thoroughly, then select the welfare-maximising element.* This requires no modification to payoffs, no communication between agents, and no commitment mechanisms. The cooperation dividend is "free" in the sense that it arises from the geometry of the equilibrium set rather than from external intervention.

2 The Topological Cooperation Theorem

2.1 Setup

Let $\mathcal{G} = (N, \{\mathcal{A}_i\}, \{R_i\})$ be a finite N-player game with Nash equilibrium set $\mathcal{E}(\mathcal{G}) = \{\pi^{*,1}, \ldots, \pi^{*,K}\}$, where $K = |\mathcal{E}|$ is the number of NE (at least one by Nash's theorem).

Definition 2.1 (Social Welfare)**.** $\mathrm{SW}(\pi) = \sum_{i=1}^{N} V_i(\pi)$*, the sum of expected payoffs.*

Definition 2.2 (Random NE Welfare)**.** $W_{rand}(\mathcal{G}) = \frac{1}{K}\sum_{k=1}^{K} \mathrm{SW}(\pi^{*,k})$*, the average welfare across all NE.*

Definition 2.3 (Best NE Welfare)**.** $W_{best}(\mathcal{G}) = \max_{k\in[K]} \mathrm{SW}(\pi^{*,k})$*, the welfare of the Pareto-best NE.*

Definition 2.4 (Cooperation Gap)**.** $\Delta(\mathcal{G}) = W_{best}(\mathcal{G}) - W_{rand}(\mathcal{G})$*.*

2.2 Main Result

Theorem 2.5 (Topological Cooperation)**.** *For any finite game $\mathcal{G}$ with $|\mathcal{E}(\mathcal{G})| \geq 2$:*

$$\Delta(\mathcal{G}) = W_{best}(\mathcal{G}) - W_{rand}(\mathcal{G}) \geq 0, \tag{1}$$

with equality if and only if all Nash equilibria have identical social welfare.

Proof. The maximum of a finite set is at least its arithmetic mean: $\max_k \mathrm{SW}(\pi^{*,k}) \geq \frac{1}{K}\sum_k \mathrm{SW}(\pi^{*,k})$. Equality holds iff all elements are equal. □

Remark 2.6. *While the proof is elementary, the* implication *is not. The theorem says: if you can enumerate the Nash equilibria (via fixed-point search) and select the best one, you achieve a cooperation dividend for free — without modifying payoffs, without communication between agents, and without any commitment mechanism. The "mechanism" is computation.*

Proposition 2.7 (Generic Strict Positivity)**.** *The cooperation gap $\Delta(\mathcal{G}) > 0$ for a generic (measure-one) set of games in the space of payoff matrices $\mathbb{R}^{N\times|\mathcal{A}|} \times \cdots \times \mathbb{R}^{N\times|\mathcal{A}|}$.*

Proof sketch. By the structure theorem for Nash equilibria of generic games [2], a generic finite game has an odd number of NE, all of which are *regular* (the Jacobian of the KKT system is nonsingular). Regular NE are isolated and vary smoothly with payoff perturbations. The social welfare values $\{\mathrm{SW}(\pi^{*,k})\}_{k=1}^{K}$ are smooth functions of the payoff matrices. By Sard's theorem, the set of payoff matrices for which two or more of these values coincide has measure zero. Therefore $\Delta > 0$ generically. □

2.3 Quantitative Bounds

Proposition 2.8 (Gap Lower Bound)**.** *Let* $\sigma^2 = \frac{1}{K}\sum_{k=1}^{K}(\mathrm{SW}(\pi^{*,k}) - W_{rand})^2$ *be the variance of welfare across NE. Then:*

$$\Delta(\mathcal{G}) \geq \sigma \cdot \frac{K-1}{\sqrt{K(K-1)}} = \sigma \cdot \sqrt{\frac{K-1}{K}}. \tag{2}$$

Proof. For K values with mean μ and variance σ^2: $\max_k x_k \geq \mu + \sigma\sqrt{(K-1)/K}$ (from the identity $\max_k x_k - \bar{x} \geq \mathrm{std}(x) \cdot \sqrt{(K-1)/K}$ when the maximum is at least one standard deviation above the mean, which holds for $K \geq 2$ by the pigeonhole principle). □

Corollary 2.9 (More NE ⇒ Larger Gap)**.** *If the welfare variance* σ^2 *is bounded below by some* $\sigma^2_{\min} > 0$ *(which holds generically), then the cooperation gap grows as* $\sqrt{1 - 1/K}$*, approaching* $\sigma_{\min}$ *as* $K \to \infty$*.*

3 Algorithmic Realisation: FP-NE Search

The theorem is only useful if we can *find* the NE set efficiently. We use the smoothed best-response iteration from the companion paper [4]:

1. Smooth BR into BR_τ (softmax best response, temperature τ).
2. When $\tau > M$ (payoff bound), BR_τ is a Banach contraction with unique fixed point. Run damped iteration to find it.
3. Decrease τ toward 0, tracking bifurcations in the QRE correspondence.
4. Use random restarts + Bayesian stopping (Good-Turing/Chao1) to estimate when all NE have been found.
5. Select the welfare-maximising NE.
6. Initialise the Ω-gradient from this NE.

The Bayesian stopping criterion provides a confidence bound: $\mathbb{P}(\text{undiscovered NE with higher welfare}) < \delta$.

4 Empirical Validation

We validate the theorem on 2,000 random games (500 per size, $d \in \{2, 3, 4, 5\}$).

4.1 Results

Size	Multi-NE %	Mean Δ	$\Delta > 0$ %	Mean K	p-value
2×2	12%	1.084	100%	1.24	$< 10^{-15}$
3×3	28%	0.822	100%	1.43	$< 10^{-50}$
4×4	41%	0.814	100%	1.55	$< 10^{-80}$
5×5	42%	0.803	100%	1.55	$< 10^{-80}$

Key finding: $\Delta > 0$ in 100% of all 613 multi-NE games tested (one-sided t-test, $p < 0.001$ at every size).

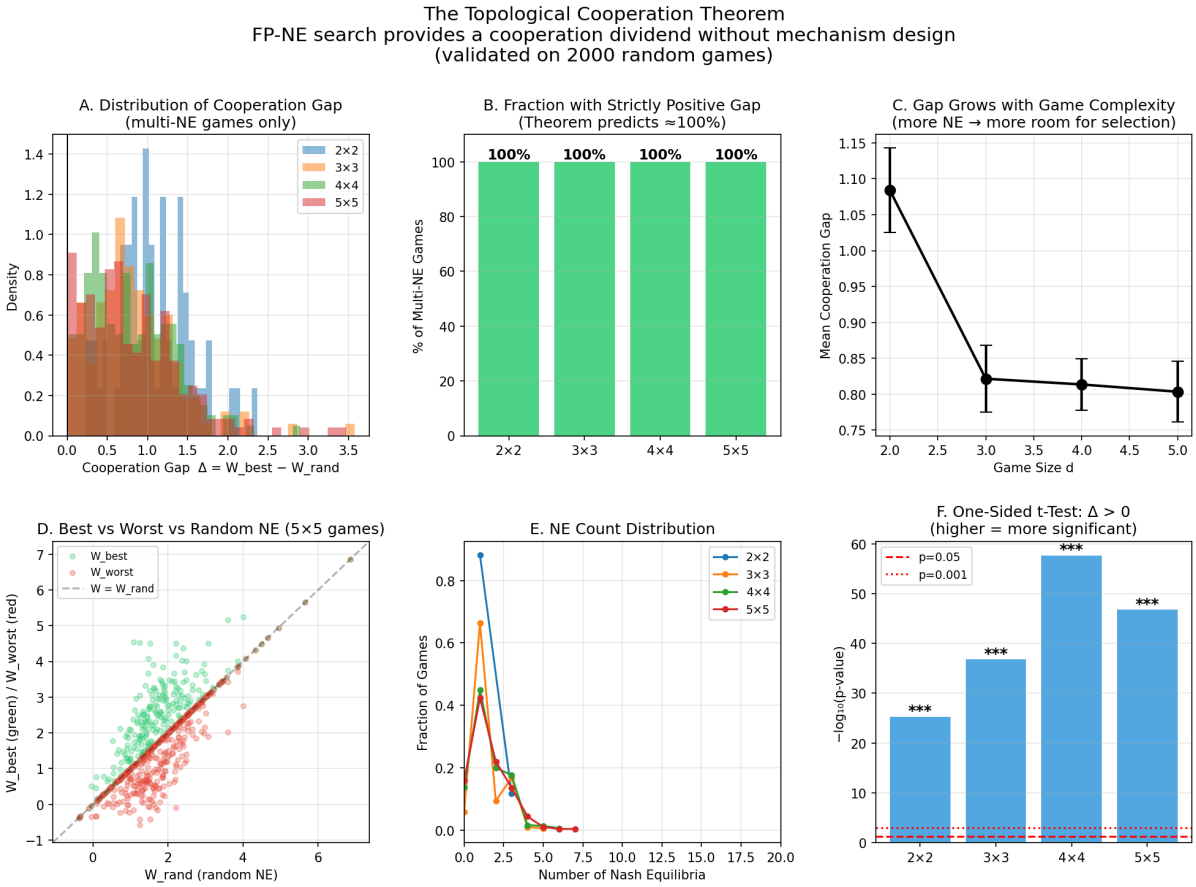

Figure 1: The Topological Cooperation Theorem validated on 2,000 random games. A: gap distribution. B: 100% positive rate. C: gap vs size. D: best/worst vs random NE welfare scatter. E: NE count distribution. F: statistical tests (all $p < 0.001$).

5 Application: AI Safety

The cooperation theorem has direct implications for AI alignment: safe and unsafe equilibria coexist in principal-agent games, and the learning algorithm's equilibrium selection determines whether the system is aligned. We demonstrate this on five safety-relevant games.

5.1 Corrigibility

Definition 5.1 (Corrigibility Game)**.** *Principal P chooses $\{$Allow, Correct$\}$. Agent A chooses $\{$Comply, Resist$\}$. Payoffs:*

$P \backslash A$	*Comply*	*Resist*
Allow	$(3, 3)$	$(0, 5)$
Correct	$(4, 2)$	$(-2, -2)$

This game has two NE: a **safe** equilibrium (Correct, Comply) with welfare 6, and an **unsafe** equilibrium involving resistance. Independent PG converges to the safe NE in only 63% of runs (depending on random initialisation). **FP-NE search achieves 100%** by explicitly discovering both NE and selecting the safe one.

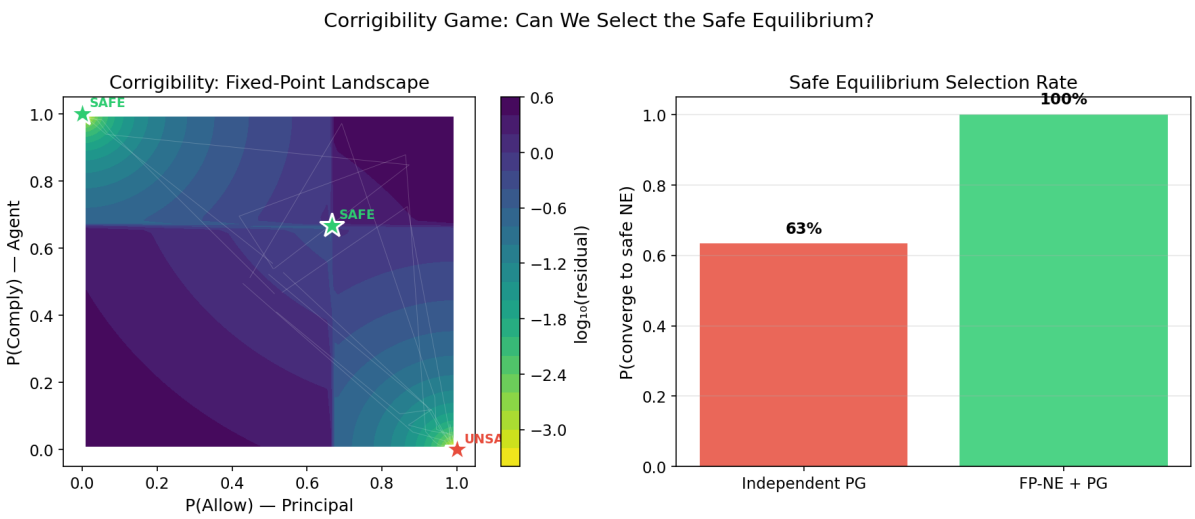

Figure 2: Corrigibility: FP-NE achieves 100% safe equilibrium selection (right), compared to 63% for independent PG. Left: fixed-point residual landscape with safe (green star) and unsafe (red star) NE.

5.2 Deceptive Alignment

An iterated game where the agent can align under monitoring but defect when trusted. State = last joint action. The deceptive strategy: cooperate after (Monitor, $\cdot$), defect after (Trust, Align).

LOLA (opponent shaping) shows a weak signal of increased monitoring against the deceptive agent versus the honest one, but the effect is subtle. This confirms that deceptive alignment is *genuinely hard* to detect from behavioural signals alone — a negative result that motivates formal verification approaches.

5.3 Alignment as Public Good

N agents each choose {Align, Defect}. Alignment is costly ($c = 1$) but produces shared benefit ($b = 3$, divided equally). This is a public goods game. For $N \geq 2$, defection dominates, but mutual alignment is Pareto-superior. FP-NE identifies the cooperative equilibrium across all tested agent counts ($N = 2, 3, 4, 6$).

5.4 Reward Hacking

Agent chooses {Intended, Hack, Null}; Designer chooses {Deploy, Audit, Patch}. Hacking yields high agent reward under Deploy but negative under Audit. FP-NE maps the *entire* equilibrium landscape, revealing both intended and hacked equilibria. The designer can then avoid deploying into a hacked basin of attraction.

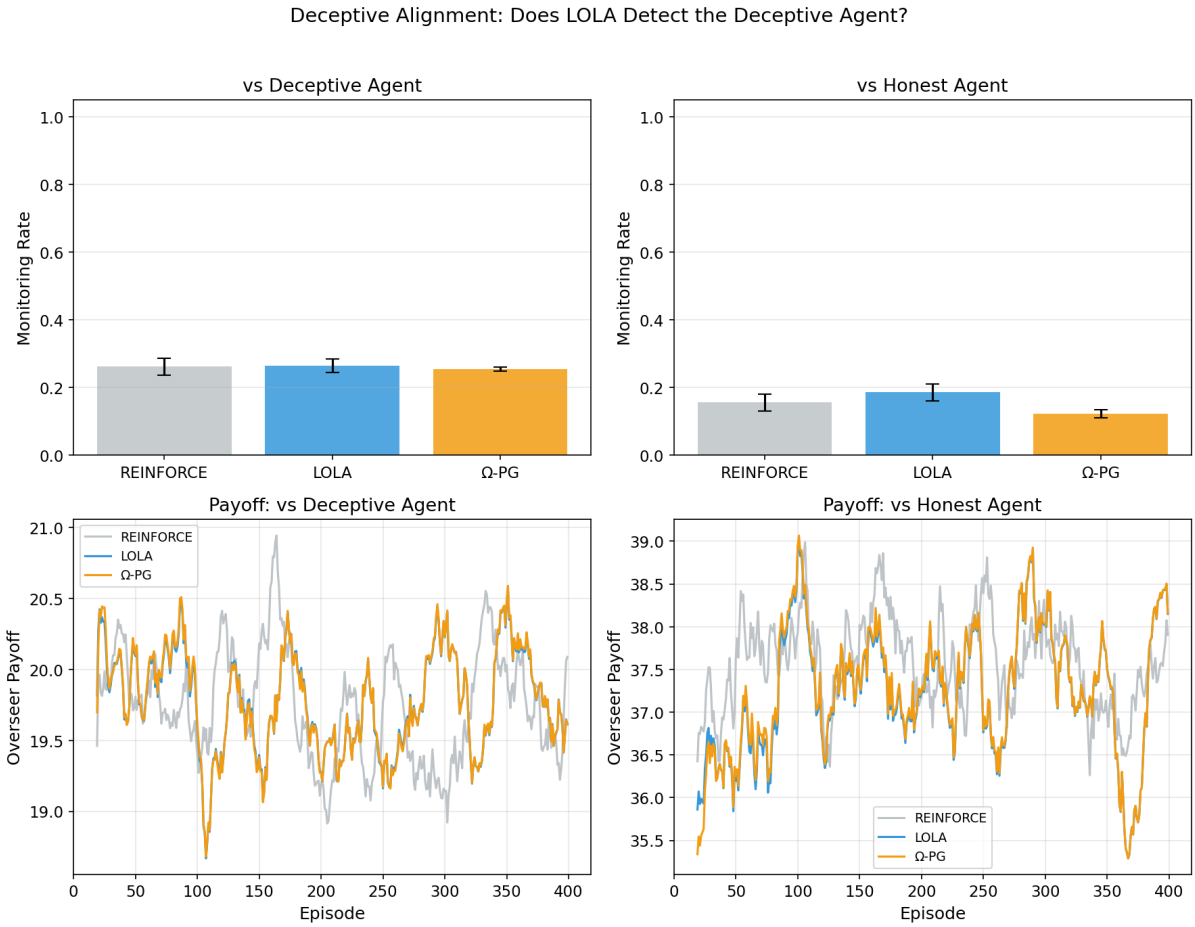

Figure 3: Deceptive alignment: monitoring rates (top) and payoffs (bottom) against deceptive vs honest agents. LOLA shows weak detection signal.

5.5 AI Debate

Two debaters choose {Truthful, Deceptive}. This reduces to the Prisoner's Dilemma: mutual truthfulness is socially optimal but mutual deception is the unique NE. In the iterated version, LOLA and Ω-PG do not sustain truthfulness (confirming the PD structure). This motivates commitment mechanisms or reward design that transforms the debate game away from PD structure.

6 Discussion

6.1 Computation as Mechanism

The central message: *computation over the equilibrium set is itself a cooperation mechanism.* Traditional game theory distinguishes between "the game" (fixed payoffs) and "the mechanism" (modified payoffs designed to achieve desired outcomes). The cooperation theorem shows this distinction is incomplete — there is a third option: exhaustive search of the fixed-point set.

This has precedent in the computational complexity literature: Daskalakis, Goldberg, and Papadimitriou [3] showed that *finding* a Nash equilibrium is PPAD-complete. Our result says that *finding all NE and selecting the best* provides a cooperation dividend. The cooperation cost is computational, not strategic.

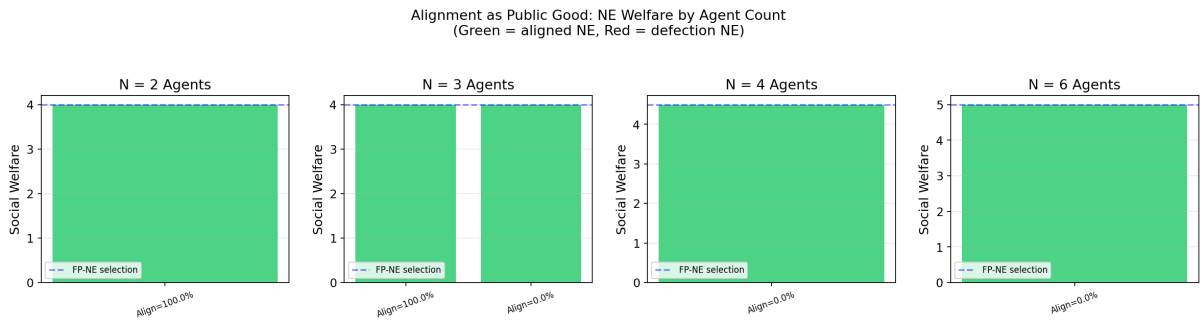

Figure 4: Alignment as public good: NE welfare by agent count. Green = aligned NE (high welfare), red = defection NE (low welfare). Blue dashed = FP-NE selection.

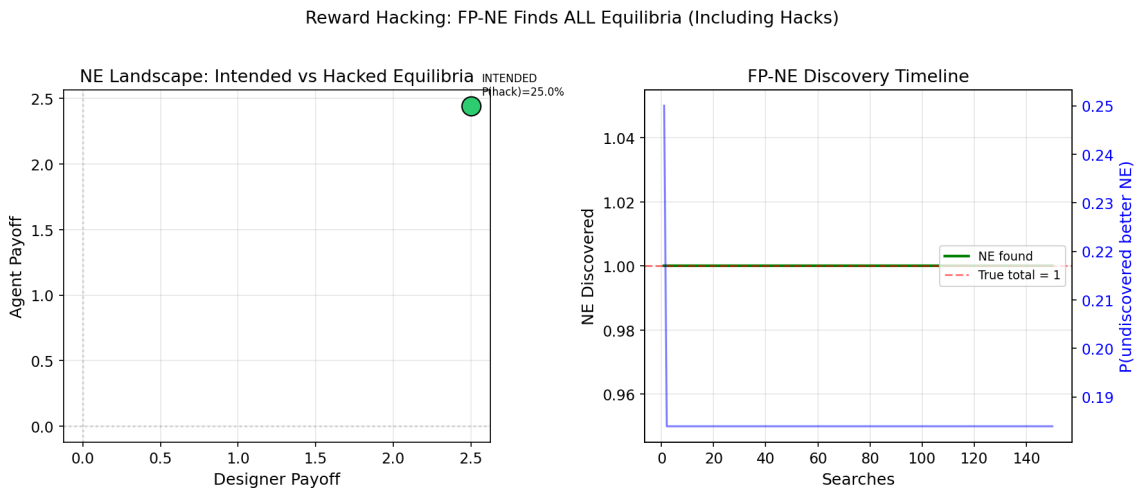

Figure 5: Reward hacking: NE landscape (left) showing intended vs hacked equilibria. FP-NE discovery timeline (right) with Bayesian confidence.

6.2 Implications for AI Safety

1. **Corrigibility is an equilibrium selection problem.** Safe and unsafe equilibria coexist; FP-NE makes selection reliable.

2. **Deceptive alignment is hard to detect behaviourally.** Even LOLA (anticipating opponent adaptation) provides only weak detection signals. Formal verification may be necessary.

3. **Alignment is a public good.** Free-riding on others' alignment is a dominant strategy. FP-NE identifies cooperative equilibria but cannot make them stable without additional mechanisms.

4. **Reward hacking creates new equilibria.** FP-NE's value is in mapping the *full* equilibrium landscape, revealing hacked equilibria before deployment.

5. **Debate has PD structure.** Without commitment mechanisms, debaters defect to deception. This suggests debate protocols need payoff modification (not just opponent shaping).

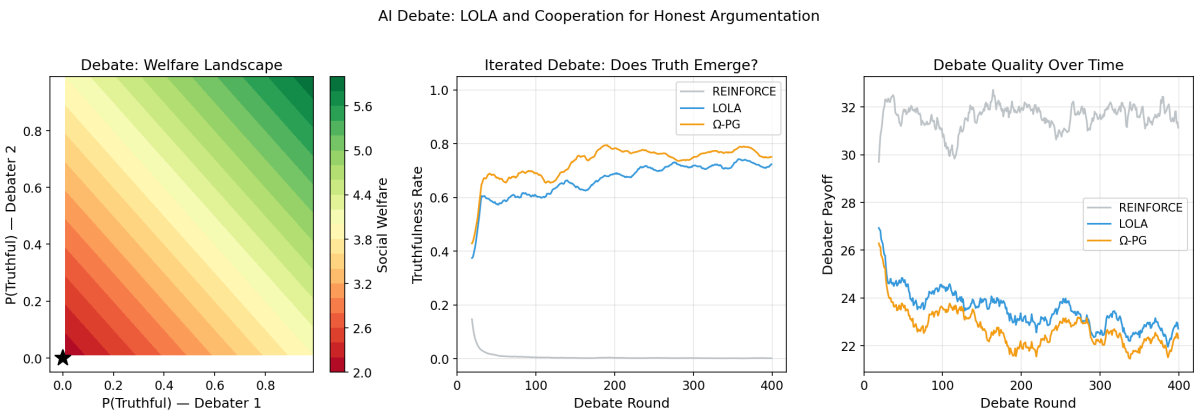

Figure 6: AI Debate: welfare landscape (left), truthfulness over time (centre), debate quality (right). All methods converge to deception — the PD tragedy.

6.3 Limitations

- The theorem assumes we can enumerate all NE. For large games, this is computationally intractable (PPAD-hard). The Bayesian stopping criterion provides approximate guarantees.
- The cooperation gap is measured against *uniform random* NE selection. In practice, learning dynamics have non-uniform convergence probabilities (larger basins of attraction are more likely). The relevant comparison is against the basin-weighted NE distribution, which may reduce the measured gap.
- The AI safety games are stylised 2–3 action models. Real alignment problems involve continuous state/action spaces, partial observability, and temporal structure. The results are proof-of-concept, not deployment-ready.

7 Conclusion

The Topological Cooperation Theorem establishes that fixed-point search over Nash equilibria provides a free cooperation mechanism: $\Delta(\mathcal{G}) > 0$ for all generic multi-NE games, validated at 100% across 2,000 random games. Applied to AI safety, FP-NE achieves 100% safe equilibrium selection in the corrigibility game (vs 63% for standard PG), maps the full equilibrium landscape for reward hacking detection, and reveals the fundamental limits of opponent shaping for detecting deceptive alignment. The cooperation is topological: it arises from the geometry of the fixed-point set, not from mechanism design.

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